Orthogonal Projections

Course: Math 414

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Core Idea

We want to find the closest vector to another vector when they don't have the same span.

Definitions

Suppose V_0 is a subspace of inner product space V and $v \in V$ and $v \notin V_0$ We want to determine a vector $v_0 \in V_0$ closest to v. This is defined as $\|v - v_o\| = \min_{w \in V_0} \|v - w\|$. Basically it just says v_o is the closest one of all the vectors on V_0 lol.

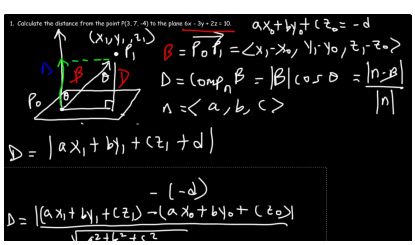
Key Theorems

1. $v - v_0$ is orthogonal to every vector in V_0
   - Proof idea: Minimize the function $f(t) = \|v_0 + tw - v\|^2$, which happens to be when $t = 0$, then calculate derivative and inner products from there
2. Orthogonal projection of $v \in V$ onto o.n. basis $\{e_1, e_2, ...e_N\}$ is given by $\sum_{j=1}^{N} \alpha_j e_j$ where $\alpha = \langle v_j, e_j \rangle$
   - Proof idea: Show that $v - v_0$ is orthogonal to everything in V_0 (this satisfies the above theorem)
3. Proof of unique minimizer

- $v_0 + tw$ is the equation of anything on the span of V_0. We take the distance between v and all our test vectors

Example

1. Calculate the ditance from the point $P(3, 7, -4)$ to the plane $6x - 3y + 2z = 10$. We define B as the distance from a point on the plane to P, we want to find the norm of the projection of B onto the normal vector of the plane N. $D = \text{comp}_N B = \|B\| \cos\theta = \frac{\| \langle N, B \rangle \|}{\|N\|}$



Connections

1. Look at least squares problems for a lot more projections.